STAT 236 Project

Square Gaussian Sparse Submatrix Detection

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1 Introduction

Sparse submatrix detection is a problem in high-dimensional statistics concerned with detecting a small signal hidden within a large noisy matrix. In gene expression data, for example, a group of genes may be elevated together in a group of samples. As with the sparse testing problems covered in this course, a central question is how the size (sparsity regime) and strength of the signal, in this case the planted submatrix, affect our ability to distinguish it from random noise.

From a purely statistical perspective, we would like to determine the weakest signal for which any test can reliably distinguish the two cases, regardless of its computational cost. However, the test that does this may require searching over an enormous number of possible submatrices. We therefore also ask how strong the signal must be for detection to be possible using an efficient algorithm, such as one running in polynomial time.

The detection problem was studied by Butucea and Ingster (2013) [1], who established asymptotic detection boundaries for the Gaussian model, and later by Ma and Wu (2015) [2], who investigated its computational limits. For recovering the affected rows and columns, we draw on Butucea, Ingster and Suslina [3] and Cai, Liang and Rakhlin [4]. These papers work in considerable generality, so we restrict to square matrices with square planted submatrices and a common positive mean shift. Note that this change required altering and thus simplifying some of the forms of the results in the original papers, such as plugging in the same values for $\{m, n\}$ or $\{k_m, k_n\}$ in cases where rectangular matrices are considered.

We introduce the three main tests used in the literature for detection: the sum of all entries, the scan over all candidate blocks, and the maximum entry. Roughly, the sum can detect weaker signals for sufficiently large blocks, while scanning helps when the block is smaller. However, scanning does become very computationally expensive. Ma and Wu show that, under a widely believed assumption known as the planted-clique hypothesis, no fast algorithm can match the scan for small blocks. In Section 3, we turn to recovering the affected rows and columns, comparing exhaustive search with spectral methods.

These comparisons concern asymptotic regimes, where the matrix size grows and the block size and signal strength may change with it. However, they do not necessarily tell us which procedure performs best on data. In that regard, in Section 4, we compare the methods on simulated data to see how well these predictions describe their detection and recovery performance. We discuss the results and possible extensions in Section 5.

In Table 1, we summarize the procedures discussed in this report. Here, n is the number of rows and columns in the observed matrix, k is the number of affected rows and columns, and μ is their common mean shift. The noise entries are independent standard Gaussian variables. We assume $k \to \infty$ and $k = o(n)$. In the table, we write $k = \lfloor n^\gamma \rfloor$, where $0 < \gamma < 1$, to describe different block sizes.

Table 1. Detection and recovery procedures

Procedure	**Paper**	**Goal**	**Regime and guarantee**	**Cost**
Sum	Butucea and Ingster [1]	Detection	Optimal for $\gamma > 2/3$. Reliable when $\mu k^2/n \to \infty$.	$O(n^2)$
Scan	Butucea and Ingster [1]	Detection	Attains the statistical detection boundary for $\gamma < 2/3$. Reliable when $\liminf \mu^2 k/[4\log(n/k)] > 1$.	$O\left(k^2\binom{n}{k}^2\right)$; needs k
Maximum entry	Ma and Wu [2]	Detection	Useful when $\gamma < 1/2$, where it needs a weaker signal than the sum. Reliable when $\liminf \mu/(2\sqrt{\log n}) > 1$.	$O(n^2)$
Sum and scan	Butucea and Ingster [1]	Detection	Attains the statistical detection boundary across both sparsity regimes.	$O\left(k^2\binom{n}{k}^2\right)$; needs k
Maximizing block sum	Butucea, Ingster and Suslina [3]	Recovery	Attains the sharp statistical boundary for exact recovery given in Section 3.1.	$O\left(k^2\binom{n}{k}^2\right)$; needs k
Spectral projection	Cai, Liang and Rakhlin [4]	Recovery	For $k/\sqrt{n} \to \infty$. Succeeds when $\mu \geq C(\sqrt{n}/k + \sqrt{\log n/k})$.	$O(rn^2)$
Thresholded spectral projection	Cai, Liang and Rakhlin [4]	Recovery	For $k/\sqrt{n} \to 0$. Succeeds when $\mu/\sqrt{\log(n/k^2)} \to \infty$.	$O(rn^2)$

For the spectral methods, r denotes the number of power iterations used to approximate the leading singular vectors. Computing a full SVD instead takes $O(n^3)$ operations. Here, C is a sufficiently large constant.

2 Detection

2.1 The Model and Testing Error

We begin with the standard Gaussian formulation of the problem. We observe a matrix $X \in \mathbb{R}^{n \times n}$ and consider the null hypothesis

$$H_0 : \quad X_{ij} \overset{\text{iid}}{\sim} N(0,1)$$

against the alternative

$$H_1 : \quad X_{ij} \overset{\text{ind}}{\sim} N\left(\mu \mathbf{1}_{\{i \in S,\, j \in T\}}, 1\right)$$

where $[n] = \{1, \ldots, n\}$ and $S, T \subseteq [n]$ are unknown subsets with $|S| = |T| = k$, and where the signal strength is $\mu > 0$. Thus, under the alternative, there is an unknown $k \times k$ submatrix whose entries have mean μ, while all other entries have mean zero.

Our goal is to understand how the difficulty of detection changes as the matrix becomes larger. As n increases, the block size and signal strength may also change, so we write $k = k_n$ and $\mu = \mu_n$. For instance, a larger block may allow us to detect a weaker signal because more affected entries contribute evidence for the alternative. We seek conditions on n, k, and μ under which the testing errors tend to zero. Throughout the asymptotic results, we assume $k \to \infty$ and $k = o(n)$.

Let φ_n be a test that takes value one when we reject H_0 and zero otherwise. We define the total testing error as

$$R_n(\varphi_n) = P_0(\varphi_n = 1) + \sup_{\substack{S,T \subseteq [n] \\ |S| = |T| = k}} P_{S,T,\mu}(\varphi_n = 0)$$

where P_0 is the null distribution and $P_{S,T,\mu}$ is the alternative distribution with affected rows S and columns T. Reliable detection is achieved when $R_n(\varphi_n) \to 0$.

2.2 The Sum Test

The first and most natural approach is to sum all entries in the matrix and compare the result with what is expected under H_0. As only detection is required, such a method is able to aggregate the effects of individually small mean shifts without identifying the specific indices. We define

$$T_{\text{sum}} = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n} X_{ij}$$

The unnormalized sum has variance n^2 under both hypotheses and mean μk^2 under the alternative. Thus, this statistic is $N(0,1)$ under H_0 and $N(\Delta_n, 1)$ under the alternative, where $\Delta_n = \mu k^2 / n$ is the mean shift of the normalized sum. We reject when $T_{\text{sum}} > c_n$, where c_n is the rejection threshold. Theorem 2.1 of Butucea and Ingster [1] shows that reliable detection is possible when $\Delta_n \to \infty$. We can then choose c_n to tend to infinity while staying below a fixed fraction of Δ_n, so that both errors tend to zero. If instead Δ_n stays

bounded, then there is no choice of c_n that makes both errors vanish. We go through their error calculation in Appendix A.1.

Computing the sum takes $O(n^2)$ operations, which is linear in the number of observed entries, and only $O(1)$ additional storage if the entries are processed sequentially. Its limitation is statistical in the sense that the unnormalized sum collects a mean shift of μk^2, but also noise with standard deviation n. For sufficiently small blocks, the many unaffected entries can obscure the signal. The sum also does not exploit the shared row and column structure, which motivates procedures that look at so-called candidate submatrices instead.

2.3 The Scan Test

A natural alternative is to sum only over a candidate submatrix. Since the affected rows and columns are unknown, we consider all possible candidates and reject if any candidate has a sufficiently large sum. For each candidate with row set A and column set B, both of size k, let $Z_{A,B} = k^{-1} \sum_{i \in A, j \in B} X_{ij}$ be its normalized sum. There are $M_n = \binom{n}{k}^2$ candidates. The scan statistic is the largest of their normalized sums

$$T_{\text{scan}} = \max_{\substack{A,B \subseteq [n] \\ |A|=|B|=k}} Z_{A,B}$$

Each candidate sum is standard normal under H_0, while the true block satisfies $Z_{S,T} \sim N(\mu k, 1)$ under H_1. However, searching over many candidates makes it more likely that some sum will be large purely because of noise. Thus, it is intuitive that the threshold must account for this search.

Let h_n denote the rejection threshold, so that we reject H_0 when $T_{\text{scan}} > h_n$. Butucea and Ingster [1, Theorem 2.1] choose $h_n = \sqrt{2 \log M_n}$. This threshold grows with the number of candidates to control false positives from the search. What is interesting is that it is approximately the largest value attained by M_n independent standard normal variables. With this choice, they show that reliable detection is possible when

$$\liminf_{n \to \infty} \frac{\mu^2 k}{4 \log(n/k)} > 1$$

Thus, it is sufficient for the entrywise mean shift μ to exceed $2\sqrt{\log(n/k)/k}$ by some fixed factor. We go through their proof in Appendix A.2, including the control of the maximum over dependent candidate sums.

One of the main issues with this test is that the scan requires knowledge of k, and this enumeration process requires us to look at M_n candidates. Computing each sum separately takes $O(k^2 M_n)$ operations, which is superpolynomial under our assumptions on k, so this better statistical guarantee comes at a substantial computational cost.

2.4 The Maximum-Entry Test

Another natural approach is related to some of the problems discussed in the course, where we considered minimum-p tests. A minimum-p test looks for a single unusually significant

observation. In our setting, this corresponds to testing whether any matrix entry is unusually large. Accordingly, define

$$T_{\max} = \max_{1 \le i,j \le n} X_{ij}$$

Let Φ denote the standard normal distribution function. Since the one-sided p-value for each entry is $p_{ij} = 1 - \Phi(X_{ij})$, we have $\min_{i,j} p_{ij} = 1 - \Phi(T_{\max})$. Higher criticism could also be applied to the n^2 entries, which is also considered in Cai, Liang and Rakhlin [4].

For this test, let u_n denote the rejection threshold, so that we reject when $T_{\max} > u_n$. Ma and Wu [2, Lemma 1] choose $u_n = \sqrt{(4+\varepsilon)\log n}$, where $\varepsilon > 0$ is a fixed constant. The largest of the n^2 null entries is approximately $\sqrt{2\log n^2} = 2\sqrt{\log n}$, so u_n lies slightly above this. In their Theorem 2, condition (17), they give the sufficient condition

$$\liminf_{n\to\infty} \frac{\mu}{2\sqrt{\log n}} > 1$$

Taking ε sufficiently small then makes the total error tend to zero. We go through their bound in Appendix A.3.

Like the sum, the maximum can be computed in $O(n^2)$ operations with $O(1)$ additional storage, and it requires neither the block size nor its location. Also like the sum, it does not exploit the shared row and column structure, since permuting all matrix entries leaves the statistic unchanged.

2.5 Comparison of Statistical Detection Regimes

The sum and scan tests use the signal in different ways. The sum combines all k^2 positive mean shifts, so a sufficiently large block can be detected even when its individual entries are only slightly elevated. For a smaller block, however, summing the entire matrix adds many unaffected entries. The scan concentrates on candidate blocks instead, but must use a higher threshold because it searches over many possible locations. Butucea and Ingster [1, Theorem 2.1] combine these approaches by rejecting whenever either test rejects. The combined test therefore succeeds whenever either condition above holds.

We can also ask whether a different test could detect substantially weaker signals. To address this, let $R_n^\star = \inf_\varphi R_n(\varphi)$ denote the smallest total testing error over all tests based on the observed matrix. Under our assumptions $k \to \infty$ and $k = o(n)$, Theorem 2.2 of Butucea and Ingster tells us that

$$\frac{\mu k^2}{n} \longrightarrow 0 \qquad \text{and} \qquad \limsup_{n\to\infty} \frac{\mu^2 k}{4\log(n/k)} < 1 \quad \Longrightarrow \quad R_n^\star \longrightarrow 1$$

Thus, if the total signal is negligible and the signal within a block is below the scan boundary, there is no test can make both errors vanish.

The theorem's additional assumptions (2.6) and (2.7) also hold in the formulation we are considering. In the paper's notation, the sampling fractions become $p = q = k/n$, so (2.7) holds trivially, and both log-log conditions in (2.6) reduce to

$$\frac{\log\log(n/k)}{\log(n/k)} \longrightarrow 0$$

which holds because $n/k \to \infty$.

We can also summarize the detection regimes by considering block sizes and signal strengths that change polynomially with the matrix size, such as in Ma and Wu [2, Section 1.2]. In our notation, we write $k = \lfloor n^\gamma \rfloor$ and $\mu = n^{-\beta}$, where $0 < \gamma, \beta < 1$. Increasing γ gives us a larger block, while increasing β makes the signal decrease more quickly. Comparing the tests therefore is similar to asking how large β can be before each test loses the ability to detect the block.

For the sum, the standardized mean shift is $\mu k^2/n = n^{2\gamma-1-\beta}(1+o(1))$, which diverges when $\beta < 2\gamma - 1$. For the scan, reliable detection occurs when $\beta < \gamma/2$. So, since we can reject using either test, we take the larger of these two limits. Together with the lower bound above, this gives the boundary

$$\beta_{\text{stat}} = \max\left\{\frac{\gamma}{2},\, 2\gamma - 1\right\}$$

Ma and Wu [2, Section 1.2], which they denote β^*. Reliable detection is possible when $\beta < \beta_{\text{stat}}$ and impossible when $\beta > \beta_{\text{stat}}$.

The two expressions agree at $\gamma = 2/3$. For $\gamma > 2/3$, the sum can detect signals that decrease more quickly than those covered by the scan condition. There are enough affected entries that adding their small positive shifts across the entire matrix is sufficient. Searching for their location does not improve the detection exponent. Ma and Wu call this the moderately sparse regime.

For $\gamma < 2/3$, the scan can detect weaker signals than the sum. The block now occupies a smaller part of the matrix, so summing everything includes too much unaffected noise. Looking within candidate blocks avoids enough of that noise to compensate for the higher threshold required by the search. This is the highly sparse regime.

Note that the max test stll has its uses here, as we will see in the simulations. When $\gamma < 1/2$, the sum requires that $\mu/n^{1-2\gamma} \to \infty$, so the individual shifts must grow polynomially with n. The max already succeeds when μ exceeds $2\sqrt{\log n}$ by a fixed factor [2, Theorem 2, condition (17)]. It can therefore detect signals much weaker than those required by the sum in this regime, without searching over blocks. However, the scan can still detect much weaker signals by combining the affected entries.

3 Recovery

In practical applications, recovering these indices of the signals may be necessary if we want to investigate the group further, where we require a procedure that estimates $\widehat{S}$ of the affected rows and $\widehat{T}$ of the affected columns. We require

$$\sup_{\substack{S,T\subseteq[n]\\|S|=|T|=k}} P_{S,T,\mu}\big((\widehat{S},\widehat{T}) \neq (S,T)\big) \longrightarrow 0$$

which is the condition for exact recovery.

3.1 Exact Recovery by Scanning

Butucea, Ingster and Suslina [3] study this problem using the estimator

$$(\widehat{S}, \widehat{T}) = \operatorname*{arg\,max}_{\substack{A,B \subseteq [n] \\ |A|=|B|=k}} \sum_{i \in A, j \in B} X_{ij}$$

This is the same search used by the scan test, but now we retain the maximizing indices.

For $k \to \infty$ and $k = o(n)$, their Theorems 2.1 and 2.3 (note this is 2.2 in the arxiv version) give the sharp recovery boundary of

$$\mu^{\star}_{\mathrm{rec}} \sim \max \left\{ \frac{\sqrt{2 \log k} + \sqrt{2 \log(n-k)}}{\sqrt{k}},\ 2\sqrt{\frac{\log(n/k)}{k}} \right\}$$

such that some fixed multiplicative separation above this boundary permits exact recovery, while a fixed separation below it makes the minimax probability of an incorrect recovery go to one. The exhaustive estimator has the same computational difficulty as the scan test, so in a similar fashion to the previous section, we next consider an efficient alternative.

3.2 Spectral Recovery

Cai, Liang and Rakhlin [4] approach recovery using the singular vectors of the observed matrix. The mean matrix $M = \mu \mathbf{1}_S \mathbf{1}_T^{\top}$ has rank one, where $\mathbf{1}_S$ and $\mathbf{1}_T$ are the indicator vectors of the affected rows and columns. The left and right singular vectors of M are proportional to these vectors.

Their Algorithm 1 computes the leading left singular vector $\widehat{u}$ and leading right singular vector $\widehat{v}$ of X. It then forms the column scores $\widehat{u}^{\top} X_{\cdot j}$ and row scores $X_{i \cdot} \widehat{v}$. Under their signal condition, the affected and unaffected indices form two separated groups of scores, and a threshold between the groups recovers the index sets.

For square blocks with $k/\sqrt{n} \to \infty$, their Lemma 1 gives a sufficiently large absolute constant C such that exact recovery succeeds whenever

$$\mu \geq C \left(\frac{\sqrt{n}}{k} + \sqrt{\frac{\log n}{k}} \right)$$

The authors suggest iterative power methods for the leading singular vectors in order to avoid the search over block locations.

3.3 Thresholding for Smaller Blocks

When $k/\sqrt{n} \to 0$, the affected block occupies a small part of the matrix, and the leading singular vectors can be dominated by noise. Cai, Liang and Rakhlin [4] therefore modify the spectral method by filtering the entries before computing the singular vectors, a technique similar to others we have seen when discussing spectral clustering.

Their Algorithm 2 uses soft thresholding at some $t > 0$, replacing each entry x by

$$\eta_t(x) = \text{sign}(x)(|x| - t)_+$$

where $(a)_+ = \max\{a, 0\}$. Entries with magnitude at most t become zero, while the remaining entries are reduced in magnitude by t. The spectral projection and grouping steps are then applied to this thresholded matrix.

The idea behind this procedure is to remove many of the small noise entries before trying to estimate the signal directions. If the affected entries have sufficiently large positive shifts, enough of their signal remains for the spectral method to identify the shared rows and columns. However, thresholding also removes affected entries when their observed values are small. We therefore cannot expect it to improve recovery in every setting, particularly when the block contains many weak individual shifts.

For this small-block regime, their Theorem 4 gives exact recovery when

$$\frac{\mu}{\sqrt{\log(n/k^2)}} \longrightarrow \infty$$

using a threshold t of order $\sqrt{\log(n/k^2)}$. This condition requires the signal to be sufficiently strong relative to the thresholding level.

In Section 4, we compare this procedure with ordinary spectral recovery for the configurations satisfying $k^2 < n$. Our implementation uses $t = \sqrt{\log(n/k^2)}$. Comparing the two methods at the same (n, k, μ) lets us examine whether removing noise compensates for the signal lost through thresholding.

4 Simulations

4.1 Setup

We use simulated matrices to compare how much signal the methods need for detection and recovery. In the first experiment, we increase μ and measure how often each test detects the planted block. In the second, we measure how often the recovery methods identify all of its rows and columns correctly. The code for all simulations is available in this Google Colab notebook.

Each matrix has independent standard Gaussian noise, with μ added to randomly chosen sets of k rows and columns. We use $n = 16$ with $k = 3, 5, 8$ and $n = 512$ with $k = 12, 42, 107$, which were sizes suggested by colab so that at each matrix size one block falls below $\sqrt{n}$, one between $\sqrt{n}$ and $n^{2/3}$, and one above $n^{2/3}$.

The detection tests use level .05. The sum and max thresholds come from their exact null distributions. We compute the scan exactly by going through all sets of k rows and taking the k columns with the largest sums over those rows, which is only feasible at $n = 16$. Its threshold is the 95th percentile of the statistic over 1,500 noise matrices.

All recovery methods are given k. Exhaustive recovery uses the same search as the scan, so it is also restricted to $n = 16$. The spectral methods take the k largest row and column scores, rather than thresholding them as in Cai, Liang and Rakhlin [4]. For soft thresholding, once again note that we use $t = \sqrt{\log(n/k^2)}$ when $k^2 < n$.

4.2 Detection

For each (n, k), we choose six values of μ that give the sum theoretical powers from .05 to .99. The dashed curve in the figure is this theoretical power, $1 - \Phi(z_{.95} - \mu k^2/n)$, which the simulated power of the sum follow closely. At $\mu = 0$, all tests reject close to 5% of the time.

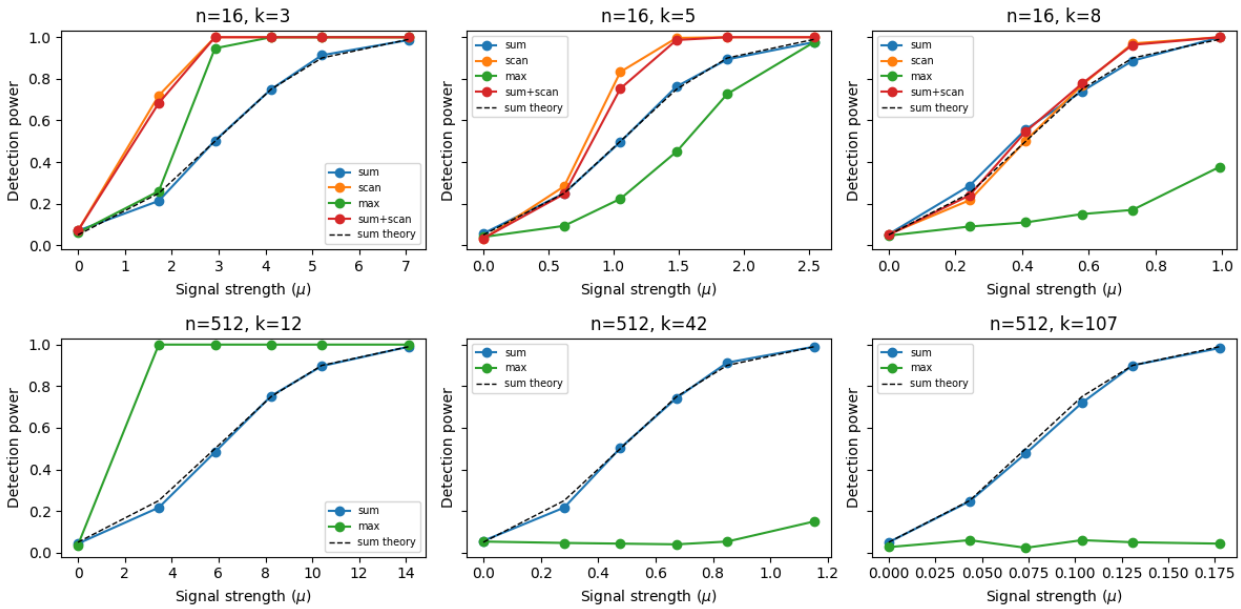


Figure 1. Detection power over 300 trials at each signal strength. The dashed curves give the exact power of the sum test.

For small blocks, scanning helps by concentrating on the affected entries. At $n = 16$ and $k = 3$, the scan has power around .7 at $\mu \approx 1.7$, while the sum is near .2. At $k = 8$, the two are close, since the block now contributes enough total shift for the sum to work. The combined test generally follows the stronger of the two.

At $n = 512$, the max works well for the smallest block, where individual shifts are large, but does little for $k = 42$ and $k = 107$. There the sum detects many weak shifts that produce no especially large entry. This switch between $k = 12$ and $k = 42$ is consistent with the crossover near $k = \sqrt{n}$ in Section 2.5

4.3 Recovery

For each (n, k), we choose signals between half and twice the recovery boundary in Section 3.1, marked by the dashed lines in the figures. Since this boundary is asymptotic, we

use it as a reference (in the sense that we are not expecting a drop exactly at the dashed line). Still, at $n = 16$ the exhaustive error drops to around .2 at that value.

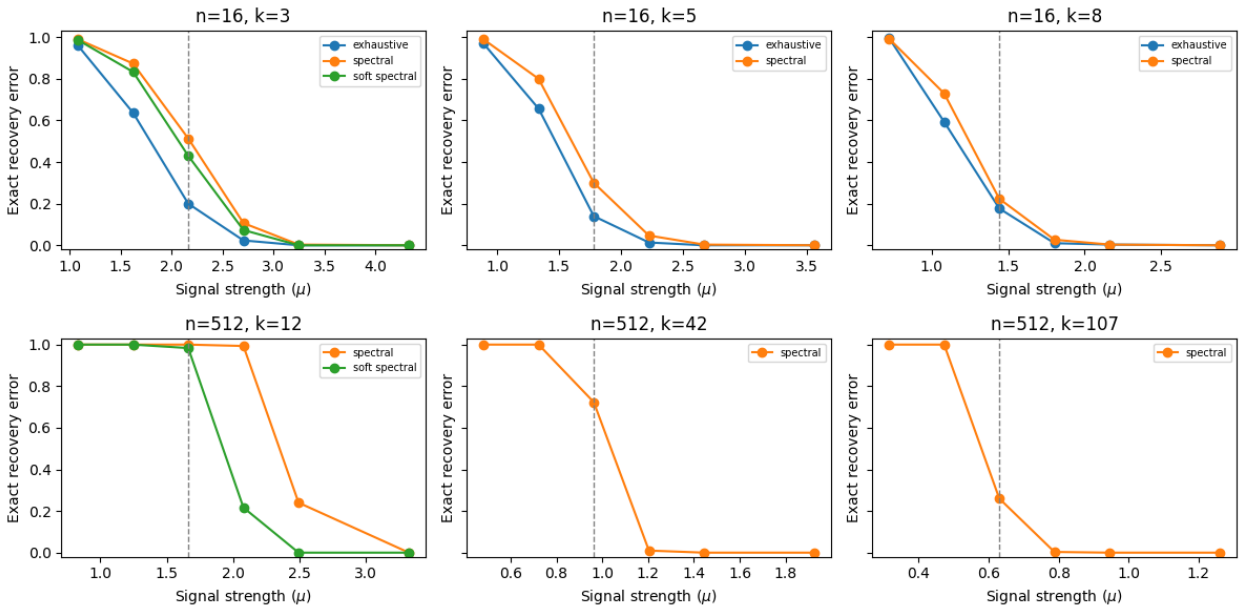


Figure 2. Exact recovery error over 300 trials at each signal strength. The dashed lines mark the asymptotic recovery reference.

Exhaustive recovery generally makes fewer errors than spectral recovery at intermediate signals. For example, at the reference with $n = 16$ and $k = 3$, their errors are roughly .2 and .5. Recovery also generally needs a stronger signal than detection. At $n = 512$ and $k = 107$, the sum detects the block almost every time once μ is about .2, but spectral recovery still fails at μ around .5. Soft thresholding helps for the smaller block at $n = 512$ and $k = 12$. Near $\mu = 2$, it cuts the error from almost 1 to about .22.

5 Discussion

Overall, the simulations show the patterns predicted by the theory in some sense. The sum detects many weak shifts, scanning helps for small blocks but can only be computed for small matrices, and the max is useful when there are individual entries that stand out in some small submatrix. Detecting a block is much easier than recovering it exactly, and soft thresholding helps spectral recovery for small blocks. These results do have some limitations, since exact scanning and exhaustive recovery were only possible at $n = 16$, which is far from the asymptotic setting. This is also true in general for the other simulations, but it is still interesting to see nonetheless.

There are also several extensions of the problem that we did not cover. We restricted to square matrices and square blocks, but Butucea and Ingster [1] and Butucea, Ingster and Suslina [3] allow rectangular ones. Their boundaries then depend on the row and column dimensions separately, and for recovery the rows and the columns each contribute their own

term. Butucea and Ingster also consider the case where the block size is only known to lie in some given set, as opposed to knowing it exactly as we have here. They also treat unknown noise variance, non-Gaussian data from an exponential family, and blocks whose entries can be shifted in either direction. Since positive and negative shifts could cancel in the sum, it would make sense that tests for that case should use squared entries instead. Finally, if the affected rows and columns must be consecutive, there are far fewer possible locations, and detection becomes easier.

It would also be interesting to think about the scan algorithmically. The exact scan searches over every set of k rows, which is what makes it so expensive. We discussed such ideas in class for vertex hunting, where there are techniques that use the geometry of the problem instead of trying each combination of vertices. A similar idea for the scan would be to build the block step by step, or to alternate between choosing the best rows for the current columns and the best columns for the current rows, as in the heuristic search that Butucea and Ingster use in their simulations. The results of Ma and Wu suggest that no efficient procedure can match the exact scan for small blocks, so it would be an interesting question as to how close such a procedure comes in practice, especially at matrix sizes where the scan is out of reach, given its statistical optimality (as a standalone or used in combination with the sum, depending on the regimes discused previosly).

A Detection Calculations

Some notes on the notation to maintain consistency. This was part of the original focus of the first draft, but after some feedback from Prof Ke, we decided to move these here instead of removing them. These sum and scan calculations follow Appendix A.1 of Butucea and Ingster [1]. Their large matrix dimensions N, M are both n here, their submatrix dimensions are both k, and their parameter a is our μ. For Ma and Wu [2], we make the changes $p = n$ and $\lambda = \mu$.

A.1 The Sum Test

Since the entries are independent and each has variance one, the unnormalized sum has variance n^2 under both hypotheses. Dividing by n therefore gives T_{sum} variance one. Under H_1, exactly k^2 entries have mean μ, so the unnormalized sum has mean μk^2. Moreover, a linear combination of independent Gaussian variables is Gaussian. Consequently,

$$T_{\text{sum}} \sim \begin{cases} N(0,1), & H_0 \\ N(\Delta_n, 1), & H_1 \end{cases} \qquad \Delta_n = \frac{\mu k^2}{n}$$

Let c_n denote the rejection threshold, so that we reject H_0 whenever $T_{\text{sum}} > c_n$. The two errors are

$$P_0(T_{\text{sum}} > c_n) = \Phi(-c_n) \qquad \sup_{S,T} P_{S,T,\mu}(T_{\text{sum}} \le c_n) = \Phi(c_n - \Delta_n)$$

Following their threshold choice, suppose $\Delta_n \to \infty$ and take $c_n \to \infty$ with $c_n \le \eta \Delta_n$ for a fixed $0 < \eta < 1$. Then the Type I error tends to zero and

$$\sup_{S,T} P_{S,T,\mu}(T_{\text{sum}} \le c_n) \le \Phi\big(-(1-\eta)\Delta_n\big) \longrightarrow 0$$

Thus, the sum test gives reliable detection when $\mu k^2/n \to \infty$.

A.2 The Scan Test

For fixed A, B with $|A| = |B| = k$, the unnormalized candidate sum has variance k^2. Hence $Z_{A,B} \sim N(0,1)$ under H_0. Under an alternative, the true candidate satisfies $Z_{S,T} \sim N(\mu k, 1)$.

For any proposed rejection threshold $u > 0$, we can union bound to get

$$P_0(T_{\text{scan}} > u) \le \sum_{\substack{A,B \subseteq [n] \\ |A|=|B|=k}} P_0(Z_{A,B} > u) = M_n \Phi(-u)$$

Note that this does not require independence between candidate sums, which is important because overlapping submatrices share observations.

We then set the rejection threshold to $h_n = \sqrt{2 \log M_n}$, so that we reject when $T_{\text{scan}} > h_n$. Butucea and Ingster use the Gaussian tail estimate

$$\Phi(-u) \asymp \frac{e^{-u^2/2}}{u} \qquad \text{as } u \to \infty$$

Here, $\asymp$ means that the quantities are bounded above and below by fixed positive multiples of one another for sufficiently large u. Since $M_n \to \infty$, we have $h_n \to \infty$, and this means that

$$P_0(T_{\text{scan}} > h_n) \le M_n \Phi(-h_n) = O\left(\frac{M_n e^{-h_n^2/2}}{h_n}\right) = O\left(\frac{1}{h_n}\right) \longrightarrow 0$$

The final equality follows from $h_n^2/2 = \log M_n$. We next control the Type II error. Under any alternative, the maximum includes the true block, so $T_{\text{scan}} \ge Z_{S,T}$. Consequently,

$$\begin{aligned} P_{S,T,\mu}(T_{\text{scan}} \le h_n) &\le P_{S,T,\mu}(Z_{S,T} \le h_n) \\ &= \Phi(h_n - \mu k) \end{aligned}$$

The bound is independent of S, T, so it also holds for the worst-case Type II error.

Equation (A.1) of the paper gives, in the square case (after some simplifications),

$$\log M_n \sim 2k \log(n/k) \qquad h_n \sim 2\sqrt{k \log(n/k)}$$

The condition $\liminf \mu^2 k/[4 \log(n/k)] > 1$ therefore gives a fixed sort of multiplicative separation between μk and h_n. Since $h_n \to \infty$, their difference tends to infinity, so the Type II error also vanishes.

A.3 The Maximum-Entry Test

Lemma 1, equation (12), of Ma and Wu [2] states that, for $u_n = \sqrt{(4+\varepsilon)\log n}$ with fixed $\varepsilon > 0$,

$$R_n\big(\mathbf{1}_{\{T_{\max} > u_n\}}\big) \le n^{-\varepsilon/2} + \exp\left[-\frac{1}{2}(\mu - u_n)_+^2\right]$$

where $x_+ = \max\{x, 0\}$. Their supremum is over a larger class of alternatives that contains ours, so the bound also holds for R_n. Ma and Wu obtain this bound by applying their scan argument from Section 6.1 with $k = 1$. Under H_0, a union bound and the Gaussian Chernoff bound used there lead to the conclusion that

$$P_0(T_{\max} > u_n) \le n^2 \Phi(-u_n) \le n^2 e^{-u_n^2/2} = n^{-\varepsilon/2}$$

Under H_1, choose an affected entry $X_{ij} \sim N(\mu, 1)$. Since $T_{\max} \ge X_{ij}$,

$$P_{S,T,\mu}(T_{\max} \le u_n) \le P_{S,T,\mu}(X_{ij} \le u_n) = \Phi(u_n - \mu) \le \exp\left[-\frac{1}{2}(\mu - u_n)_+^2\right]$$

In Section 4.3, the authors apply it under the second condition in (17) of Theorem 2. If $\liminf \mu/(2\sqrt{\log n}) > 1$, taking ε sufficiently small makes $\liminf \mu/u_n > 1$. Since $u_n \to \infty$, this gives $\mu - u_n \to \infty$, and both error bounds tend to zero.

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